

- Q.28 What are the principles of automatic control system?
- Q.29 How does Force feedback work? Explain with loop diagram.
- Q.30 What is the role of indications in landing gear?
- Q.31 Explain redundancy and emergency.
- Q.32 What are the different types of fire protection systems?
- Q.33 How does steering system work?
- Q.34 How do flow integrators work?
- Q.35 What are the applications of anti icing and de-icing systems?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe the different types of oxygen systems and emergency systems.
- Q.37 Write in brief about
- liquid to gas converters
 - Fire Detection Systems
- Q.38 How are windows and emergency exits designed in accordance with Seat Safety instructions? Describe anti-skid system.

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187753/147753

5th Sem / AME Subject:- Aircraft Systems

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the primary function of accumulators in hydraulic systems?
- To regulate fluid flow
 - To store energy and maintain pressure
 - To filter contaminants from the fluid
 - To control the direction of fluid flow
- Q.2 Which of the following is the body's most common response to hypoxia?
- Hyperventilation
 - Hypoxia
 - Barotrauma
 - Sinus
- Q.3 The most common hydraulic fluid is:
- Mineral oil
 - Synthetic fluid
 - Water
 - Gel
- Q.4 What is the primary purpose of contamination control filters in hydraulic systems?
- To regulate fluid flow rate
 - To prevent fluid leakage
 - To remove impurities and particles from the fluid
 - To control the direction of fluid flow

- Q.5 Which of the following is not shown by backup instruments?
- a) Speed b) Altitude
c) Heading d) Passengers onboard
- Q.6 What is the disadvantage of bladder fuel tank?
- a) Loss of available fuel volume
b) Self sealing
c) No loss in fuel volume
d) Thrust loading
- Q.7 Landing gears fold away during the flight to reduce _____
- a) Drag b) Thrust
c) Airspeed d) Altitude
- Q.8 What is the disadvantage of bladder fuel tank?
- a) Loss of available fuel volume
b) Self sealing
c) No loss in fuel volume
d) Thrust loading
- Q.9 The pressure inside the cabin is technically referred to as _____
- a) Cabin bar b) Cabin pressure
c) Cabin altitude d) Cabin level
- Q.10 The air used for pressurization is usually bled off from the engines, at the _____ stage.
- a) Cryogenic b) Compressor
c) Thermal d) Hydraulic

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is reservoir of hydraulic system?
- Q.12 What do you mean by force feedback?
- Q.13 What is heat load?
- Q.14 What is the need for oxygen bottles in aircraft?
- Q.15 What is the use of a pressure suit?
- Q.16 Name the different shock absorbing devices.
- Q.17 What do you mean by frequency response?
- Q.18 Give two examples of Pilot's personal equipment.
- Q.19 What is a life raft?
- Q.20 What is the purpose of a seat belt?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What is accumulator and its use?
- Q.22 What is a servo and its use?
- Q.23 What is the use of a pneumatic system?
- Q.24 Where in the world anti icing system is required?
- Q.25 What are the components of landing gears?
- Q.26 How do pressure suits work?
- Q.27 What do you mean by flame proofing?